Homework problems:

Exercise 2.1 (2 points)**.** Answer all the questions from the Moodle Quiz **Intro to data**.

> *The quiz is available from April 20, 19:00 until April 27, 09:00. You have 45 minutes to answer the questions.*
>
> **Remark:** This quiz contributes to the grade bonus. However, its main purpose is to familiarize you with this quiz format. It is worth only two points, so the risk is minimal if the outcome is not as expected.

Solution: To see the solution to this problem, please reopen the quiz and click on *Review*.

Exercise 2.2.
A study utilizing three nationally representative, large-scale data sets from Ireland, the United States, and the United Kingdom (n = 17247) surveyed adolescents aged 12 to 15. Participants maintained a screen time diary and responded to questions regarding their feelings and behaviors. These responses were used to calculate a psychological well-being score. Additional variables, including each child's sex and age, as well as maternal education, ethnicity, psychological distress, and employment status, were incorporated into the analysis. The study found minimal evidence to suggest that screen time negatively impacts adolescent well-being.

a) Identify the type of study conducted in this research.

b) Specify the explanatory variables included in the analysis.

c) Specify the response variable measured in the study.

d) Discuss whether the study results can be generalized to the broader population, providing justification.

e) Comment on whether the study results support the establishment of causal relationships.

Solution:

a) The study is classified as observational.

b) The explanatory variables include screen time, the child's sex and age, as well as maternal education, ethnicity, psychological distress, and employment status.

c) The response variable measured is psychological well-being.

d) The results of the study can be generalized to the broader population because the data were drawn from nationally representative samples.

e) The results of the study cannot be used to establish causal relationships because the research design is observational.

Exercise 2.3.
The summary table below presents the number of space launches in the United States, categorized by launch agency type and launch outcome (success or failure).

	1957 - 1999		2000 - 2018	
	Failure	Success	Failure	Success
Private	13	295	10	562
State	281	3751	33	711
Start up	-	-	5	65

a) Identify the variables collected for each launch that were used to construct the summary table above.

b) Indicate whether each variable is numerical or categorical. For categorical variables, specify whether they are ordinal.

c) To analyze how launch success rates differ among agencies and across time periods, identify the response variable and the explanatory variables in this context.

Solution:

a) The variables collected for each launch include year, launching agency, and launch outcome (success or failure).

b) The variable `year` is an ordinal categorical variable with two levels: "1957-1999" and "2000-2018". The ordering reflects the chronological sequence of the periods. Both `launching agency` and `launch outcome` are nominal categorical variables.

c) The response variable is `launch outcome` (success or failure), while the explanatory variables are `year` and `launching agency`.

Exercise 2.4.

Complete the IDSST R tutorial `data-and-visualization` until April 27.